

Issues — Open Workable Items

Revision: 11 **Last modified:** 2026-08-18T21:33:30Z **Ticket prefix:** BOB (operator-mandated, 2026-06-06) **Scope:** Open/active items only. Closed items migrate to [Fixed.md](#).

Tracking: this file + [Issues_Summary.md](#) are authoritative for open work. The SQLite single-source-of-truth + docs_chain engine (BOB-010) is complete.

BOB-008 — RuTracker automated login blocked by CAPTCHA

Status: Operator-blocked **Type:** Bug **Created:** 2026-06-06
Operator-Block-Details: WHAT — RuTracker login with stored creds returns no session cookie (CAPTCHA wall). WHY — automated user/pass login is CAPTCHA-gated; self-resolution exhausted (creds correct + wired, login attempted, auth=True). UNBLOCK — [A] operator completes the CAPTCHA flow at /api/v1/auth/rutracker/captcha + /login. [B] operator pastes a fresh bb_session cookie via /auth/rutracker/cookie-login. WHO — operator.

Evidence: live search per-tracker stat rutracker status=error auth=True error="login returned no session cookie – likely CAPTCHA".

BOB-065 — Lava P2: Egress diagnosis and VPN-host SOCKS routing (containers pkg/egress)

Status: Queued **Type:** Task **Severity:** High

[Backfill from RD2-15/GA-05, audit doc 2026-08-08] Lava-porting finding P2 (Egress decision + VPN-host routing, Lava PLAYBOOK sections 0 and 4). Problem Boba-Base shares: on a datacenter host, trackers are network-blocked (DNS-fail/TLS-MITM, not Cloudflare so FlareSolvrr cannot fix). Affects Jackett indexer fetches + merge_service/download-proxy + plugin engines.

Diagnosis (port the script): curl https://api.ipify.org (host IP) + curl -o /dev/null -w %{http_code} https:/// direct vs via a VPN-host SOCKS proxy. Different egress IP + 200 via proxy confirms. Fix: route outbound through a VPN-connected host (the nezha pattern). SOCKS tunnel ssh -D 127.0.0.1:1080 -N (use -socks5-hostname for remote DNS); point Jackett + download-proxy + qBitTorrent-go at it (P3). For browser-cookie harvest, run the harvester ON the VPN host. Port: containers submodule pkg/egress (tunnel up/verify) + scripts/egress-via-vpn.sh glue; reuse Boba-Base existing ensure-macos-tunnel.sh style. TDD: assert the via-proxy egress IP != direct host IP AND a known-blocked tracker returns 200 via proxy. Source: docs/PORTING-FROM-LAVA.md. Per audit RD2-15 [P0]: Create tracked workable items (BOB-064..067) for the four Lava-porting findings, citing implementing commits as evidence, closed as Implemented.

BOB-066 — Lava P3: BOBA_UPSTREAM_PROXY in download- proxy + qBitTorrent-go + Jackett + compose env-forward

Status: In progress **Type:** Task **Severity:** High

[Backfill from RD2-15/GA-05, audit doc 2026-08-08] Lava-porting finding P3 (Configurable outbound proxy in the services, Lava PLAYBOOK section 3). download-proxy (Python): httpx/requests honor HTTP_PROXY/HTTPS_PROXY/ALL_PROXY/NO_PROXY env natively — add an explicit BOBA_UPSTREAM_PROXY config that sets these for tracker-bound clients, with loopback bypass (NO_PROXY=127.0.0.1,localhost,jackett). qBitTorrent-go: set http.Transport.Proxy (socks5 native, remote DNS) from a BOBA_UPSTREAM_PROXY env — mirror Lava internal/httpx/proxy.go. Jackett: has a built-in proxy setting (configure via its API/ServerConfig). Deploy gotcha (port): the env must be FORWARDED into the containers (docker-compose.yml env / the boba-ctl deploy) — Lava bug was a missing allow-list entry. Verify on distroless via podman inspect, not exec printenv. TDD: a test with a local proxy asserts the service tracker request traverses it; falsifiability: disable the wiring so the test fails. Source: docs/PORTING-FROM-LAVA.md. Per audit RD2-15 [P0]: Create tracked workable items (BOB-064..067) for the four Lava-porting findings, citing implementing commits as evidence, closed as Implemented.

BOB-068 — RD2-00: unattributed, unreviewed Auto-commit mechanism pushing to main

Status: Queued **Type:** Bug **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-00, Rev-6 downgrade to P1] Three commits at HEAD carry the bare message Auto-commit with no body, no ticket reference, no TDD trail: 54e313f (26 files), 9c8f684 (6 files), 743097a (2 files). Two landed while the audit was in progress. Source unidentified: no crontab, no systemd timer, no matching process, no in-tree script emitting the string, no hooksPath, no PostToolUse/Stop hook. Timezone +0500 on newer commits does not match host +0300 MSK. RD2-00 Update: git reflog proves 9c8f684/743097a arrived via pull: Fast-forward from a +0500 host, matching the established 2026-06-28 cross-host rsync-sync pattern (55b8671/cdb555f). Update 2 (Rev-6): root cause is a second live Claude session (Opus 5, same +0500 host) that independently landed constitution commit 177f2b0 (new anchor §11.4.238) while this session was mid-edit on the identical anchor number. Downgrades severity from P0 to P1. Every such commit bypasses §2 wrapper, §11.4.43 TDD-fix, §11.4.92 5-pass eval, §11.4.125/§11.4.142/§11.4.194/§11.4.209 mandatory Fable-xhigh review, §11.4.234 dedicated commit/push script. Composes with RD2-10..13.

BOB-069 — RD2-40: §11.4.238 QA-discovery-channel ledger — ongoing coverage-escape backfill

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md Rev-6 addendum, priority downgraded P0->P1] §11.4.238 compliance MECHANISM STOOD UP AND EXERCISED 2026-08-08. docs/QA_DISCOVERY_LEDGER.md created with discovery-channel schema, seeded with 4 real entries from this session (RD2-22, RD2-41a, RD2-41b, BOB-008), each with a real coverage-escape audit citing the specific missing/blind check; 3 of 4 already closed with a genuine new automated check carrying real RED->GREEN evidence. CLAUDE.md Critical Constraints now points at the ledger and states the discovery-channel-split mandate. Closed-loop example exercised end-to-end: scripts/pre_build_verification.sh invariant 17 extended to run workable-items diff (not just validate) — RED-captured via tests/unit/test_pre_build_workable_items_diff_check.sh §1.1 paired mutation. Honest boundary (§11.4.6): full retroactive audit of every one of

the ~50 GA-NN/RD2-NN findings in this document (each needs its own escape-audit entry) was NOT attempted — ledger is honest that it starts now (100% out-of-band in its own tracked split table) rather than backfilling a false complete history. BOB-009/BOB-010 evidence_path gap remains open. Priority: downgraded from P0-blocking to P1-ongoing — MECHANISM exists and applies to new findings; the remaining retroactive audit + BOB-009/010 closure is incremental.

BOB-070 — RD2-41: pre-build mutation-marker scan carrier false-positive silently defeats entire pre-build gate

Status: Queued **Type:** Bug **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md Rev-6 addendum, RD2-41 P1 — arguably P0] scripts/docs_chain.sh (FIXED this session) and scripts/pre_build_verification.sh invariant 17 (FIXED this session) both hardcoded a non-existent bin/workable-items path, silently no-op ing/skipping DB-export and DB-validate steps on every run — root-caused and fixed via the resolution chain proven in constitution/scripts/reporting/report_item.sh (env override -> committed constitution copy -> on-demand go build). Regression guards: tests/unit/test_docs_chain_binary_resolution.sh (RED->GREEN) + tests/unit/test_pre_build_workable_items_invariant.sh (RED->GREEN). A SEPARATE, still-open defect surfaced while fixing this: pre_build_verification.sh pre-code-review mutation-marker scan aborts the ENTIRE script (exit 1) before invariant 17 is ever reached, on carrier false-positives — legitimate constitution/**/*_mutation_test.sh + *_test.go files that intentionally contain the literal strings MUTATED/# MUTATION as part of their OWN §1.1 testing logic (36 hits, all under constitution/scripts/{gates,multitrack,workable-items}/). Same §11.4.201 carrier-false-positive class as GA-24 + RD2-01. Practical impact: the ENTIRE pre-build gate has not completed a full run for as long as this false-positive has been live — no invariant past the mutation-marker check has been genuinely enforced. Fix needed: marker-scanner to exclude self-referential test/gate files (or use structural check than bare substring), matching the fix scoped for GA-24/RD2-01/RD2-36. Priority: P1, arguably P0.

BOB-071 — RD2-01: guard-forbidden-commands.sh hook has live reproducible substring carrier false-positive

Status: Queued **Type:** Bug **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-01, P2] During this investigation the command `echo === systemd system-level (may need no sudo for list) ===` was BLOCKED by the PreToolUse hook with BLOCKED — §6.U no-sudo, because the guard does a substring match for sudo against the ENTIRE command line, including inside an unrelated echo string (need no sudo for list). Exact §11.4.201 carrier-false-positive class GA-24 already documented for a different file — now independently reproduced live, proving structural pattern in guard matching approach not a one-off content gap. Fix: guard needs word-boundary / shell-token-aware matching (or restrict sudo/su check to actual command-invocation position) rather than an unanchored substring grep across the whole line, mirroring fix direction scoped for GA-24 (EXCLUDE_PATHS is band-aid per-file; root cause is matching strategy itself). Priority: P2 (annoying, self-correcting via retry, but real false-positive class that will keep recurring). Composes with RD2-36 (canonical remediation).

BOB-074 — RD2-07: DDoS-class testing fully absent from the mandated test-type matrix

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-07, P2] tests/ has 18 subdirectories covering unit/integration/e2e/security/chaos/stress/performance/benchmark/UI (13 of the constitution ~14 mandated categories present). tests/load/locustfile.py covers throughput/scale only. No file anywhere mentions DDoS, flood, or attack-resilience testing — a distinct mandated category per §11.4.27, and not legitimately N/A given this is an internet-facing FastAPI/Gin service with public HTTP endpoints reachable over a LAN tunnel. Fix: author tests/ddos/ (or extend tests/load/) with rate-limit-enforcement, malformed-request-flood, and resource-exhaustion-under-attack coverage for the exposed download-proxy/merge-service endpoints. Priority: P2. Canonical implementation: RD2-32.

BOB-076 — RD2-09: submodules/ jackett fork 1 commit behind upstream (informational)

Status: Queued **Type:** Task **Severity:** Low

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-09, P3 informational] git submodule status + fetch shows submodules/jackett 1 commit behind origin/master (e12342eb4, a trivial false-positive fix). All other submodules (constitution, challenges, containers, helixqa) are exactly at their upstream tip. Fix: bump when convenient; not blocking anything. Priority: P3. Canonical implementation: RD2-39.

BOB-077 — RD2-10: Identify second host running the Auto-commit rsync/ sync mechanism (OPERATOR- DECISION)

Status: Operator-blocked **Type:** Task **Operator-Block-Details:** WHAT: OPERATOR must identify which second host holds push credentials + confirms whether the rsync/sync mechanism is intentional or a stale job to retire. This session cannot inspect a host it has no access to (§11.4.6/§11.4.101). WHY: This session cannot inspect or reach the +0500 host that produced the Auto-commit fast-forwards (§11.4.6 no-guessing, §11.4.101 reversible-safe default). UNBLOCK: [A] operator names the second host + confirms intentional (proceed to RD2-11 wiring) · [B] operator confirms stale/misconfigured (retire the job) · [C] operator confirms the second live Claude session/device is the source (Rev-6 Update 2 root cause, proceed to per-session discipline enforcement) WHO: Operator **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-10, P0 OPERATOR-DECISION] Root-caused as far as this host allows (see RD2-00 Update): reflog proves the two newest commits arrived via plain pull: Fast-forward from a +0500 host, matching this project established 2026-06-28 cross-host rsync-sync pattern (cdb555f/55b8671). Needs operator input to go further — which second host runs it, and is it the intended mechanism (just needs a real message + review gate) or a stale job to retire. Not auto-executed (§11.4.6/§11.4.101 — this session cannot inspect or guess at a host it has no access to). Rev-6 update: root cause narrowed to a second live Claude session (Opus 5, same +0500 host) that independently landed constitution 177f2b0. Priority: P0 (operator-blocked).

BOB-078 — RD2-11: Once identified, wire Auto-commit mechanism through §11.4.234 dedicated commit/push script OR retire it

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-11, P0] Once identified (RD2-10): either wire it through a real §11.4.234 dedicated commit/push script (with the hook-validation stages this project already has via .claude/settings.json PreToolUse guard, extended to a real pre-commit content check) or shut it down if it serves no purpose. Priority: P0. Blocked on RD2-10.

BOB-079 — RD2-12: Retroactive attributed history notes for GA-18/21/22/25/26/27 changes (never rewrite published history)

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-12, P1] Retroactively give proper, attributed, reviewed commit messages / history notes to the technically-correct changes the Auto-commit mechanism already made (GA-18, GA-21, GA-22, GA-25, GA-26, GA-27) — this can be a documentation/changelog note rather than a history rewrite (§11.4.113 — never rewrite published history), citing the git-history evidence per §11.4.124 investigate-before-attribute pattern. Priority: P1.

BOB-080 — RD2-13: Retroactive Fable-xhigh code review of the substantive Auto-commit diffs

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-13, P1] Retroactively run the mandatory independent Fable-xhigh code review (§11.4.125/§11.4.142/§11.4.209) against the substantive diffs the Auto-commit mechanism introduced (start.sh three new functions especially, since they have zero test coverage — see Root Cause 4 / RD2-24) — even though the changes already

landed, a post-hoc review closes the governance gap and will surface RD2-24 (GA-27 missing tests) if not already caught. Priority: P1.

BOB-081 — RD2-14: Author CONTINUATION.md Session 15 entry (currently 53 days / 24+ commits behind HEAD)

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-14, P0 — closes GA-04] Author a new CONTINUATION.md Session 15 entry summarizing every substantive commit since 646b295 by theme (Cyrillic/UTF-8 fix wave, tracker/plugin fixes, egress/proxy/Jackett feature wave, the two governance audits, RD2-00 discovery). Update **Last commit:/Last modified:/Revision:**. GA-04 evidence: still Revision: 19, Last modified: 2026-06-16T23:55:00Z, Session 14 — ~53 days / 24+ commits behind HEAD. Priority: P0.

BOB-082 — RD2-15: Create BOB-064..067 workable items for the four Lava-porting findings (closes GA-05)

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-15, P0 — closes GA-05] Create tracked workable items (BOB-064..067, via the workable-items tool — never raw MD edits) for the four Lava-porting findings, citing implementing commits as evidence, closed as Implemented. GA-05 evidence: `grep -in lava|BOB-06[4-7] docs/Issues.md docs/Fixed.md` returns zero hits. Priority: P0. NOTE: BOB-064..067 have been created 2026-08-10 as Task/Queued (the four Lava P1..P4 items); the closed-as-Implemented status flip (citing per-finding implementing commits) remains owed under this item.

BOB-083 — RD2-16: Regenerate browser_extension/features/codegraph Status.md + Summary/HTML/PDF siblings

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-16, P1] Regenerate docs/browser_extension/Status.md, docs/features/Status.md, docs/codegraph/Status.md + their Status_Summary.md/HTML/PDF siblings (§11.4.45/§11.4.56). Priority: P1. Composes with RD2-08.

BOB-084 — RD2-17: Reconcile BOB-008 DB/MD body drift via the workable-items tool

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-17, P1] Reconcile BOB-008 DB/MD body drift (RD2-04) via the workable-items tool. Priority: P1. Composes with RD2-04 (evidence source) and RD2-20 (preventive fix).

BOB-085 — RD2-18: Create top-level Boba proxy/merge-service v1.0.0 readiness ledger (closes GA-10)

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-18, P2 — closes GA-10] Create the top-level Boba (proxy/merge-service) v1.0.0 readiness ledger (GA-10) — dedupe with the browser_extension existing one as the template. GA-10 evidence: only docs/RELEASE_READINESS_20260616.html/.md/.pdf (dated point-in-time snapshot) and the extension own ledger exist; no top-level proxy/merge-service ledger created. Priority: P2.

BOB-086 — RD2-19: Fix BOB-009/ BOB-010 evidence_path + backfill item_history for 56 silent closures

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-19, P2] Fix BOB-009/BOB-010 evidence_path values + backfill item_history audit trails for the 56 silent closures where recoverable from git log (RD2-03). Rev-6 note: no workable-items subcommand exists to fix a historical closure evidence_path — see Root Cause 2 in the audit. Priority: P2.

BOB-087 — RD2-20: Wire docs_chain / commit-seam sync hook per §11.4.106(F) so DB writes cannot land without MD mirror

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-20, P0] Wire (or fix) the docs_chain / commit-seam sync hook per §11.4.106(F) so a docs/workable_items.db write can never again land without its MD mirror in the same commit — this is the mechanical fix that prevents Root Cause 2 from recurring, not just a one-time catch-up. Priority: P0.

BOB-088 — RD2-21: Complete/verify README Tracked-Items + Status Documents table row-completeness (GA-07 remainder)

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-21, P1] Complete/verify the README Tracked-Items + Status Documents table row-completeness (GA-07 remaining half). Not independently re-verified this round whether every mandated doc (CONTINUATION.md, Issues.md, Fixed.md, PORTING-FROM-LAVA.md, both new GA/RD2 audit docs, every Status.md/Status_Summary.md pair) actually has a row. Priority: P1.

BOB-089 — RD2-24: RED-first tests for start.sh reload_python/reload_plugins/recreate_stack (closes test-half of GA-27)

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-24, P1 TDD — closes test-coverage half of GA-27] Author RED-first tests for start.sh reload_python(), reload_plugins(), recreate_stack() — real executing tests through the actual invocation path (per §11.4.224(A): never a bash -n parse-check alone), asserting exit status + the documented side effects (pycache-clear-then-restart, restart-only, down-then-up) against a live or realistically-mocked container runtime. Verification: tests fail against a stubbed/reverted version of the three functions, pass against current start.sh (§11.4.115). GA-27 evidence: grep -rln reload_python|reload_plugins|recreate_stack tests/ challenges/ returns zero hits — shipped with no test-first coverage, a direct §11.4.224 violation. Priority: P1.

BOB-090 — RD2-25: HelixQA Challenge entry exercising all three start.sh subcommands end-to-end against real compose stack

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-25, P2] Add a HelixQA Challenge entry exercising all three start.sh subcommands (reload_python, reload_plugins, recreate_stack) end-to-end against the real compose stack. Priority: P2.

BOB-091 — RD2-26: Relocate mocked SearchOrchestrator tests to unit/ + author real-service replacements (closes GA-14/15/16)

Status: Queued **Type:** Bug **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-26, P1 — closes GA-14/15/16] For each of tests/integration/test_merge_api.py, tests/e2e/test_full_pipeline.py, tests/contract/

test_tracker_stats_contract.py: (a) relocate the existing mocked version into tests/unit/ under an honest name (the coverage is valuable as unit-level route-contract testing — keep it, just where it belongs); (b) author a real-service replacement in the original directory using the repo existing live-stack fixtures (tests/fixtures/compose.py, tests/fixtures/services.py, tests/integration/test_fixtures_bring_up_services.py already prove this pattern works here). Verification: each relocated/replaced file directory-appropriate real-service test actually issues a real HTTP call to a real running service and asserts on the real response — no @patch/monkeypatch targeting SearchOrchestrator anywhere outside tests/unit/. Priority: P1.

BOB-092 — RD2-27: Remove test_live_stack_evidence.py:265 nnmclub SKIP-on-404 fallback + verify live 200 (closes GA-13)

Status: Queued **Type:** Bug **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-27, P1 — closes GA-13] tests/e2e/test_live_stack_evidence.py:265 — with the live stack up, confirm curl :7187/api/v1/auth/nnmclub/status returns 200 (redeploy first if source≠artifact drift persists — see Root Cause 6), remove the skip fallback entirely, let the real assertion run. Priority: P1.

BOB-093 — RD2-28: Live compose bring-up + verify rutracker ReDoS regex bounds deployed to container (closes runtime half of GA-12)

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-28, P1 — closes runtime-verification half of GA-12] Bring up the live compose stack (./start.sh), run ./install-plugin.sh, verify grep {0,512} config/qBittorrent/nova3/engines/rutracker.py on the container, and capture timing of a large rutracker result page (<2s per the original RW-06 acceptance criterion). GA-12 status: DONE (source) at plugins/rutracker.py:140 confirmed {0,512}/{0,256}-bounded; runtime unverified (no live container stack running at investigation time). Priority: P1.

BOB-094 — RD2-29: Author tests/ stress/ test_tracker_fetch_stress_chaos.py with §11.4.85 fault injection

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-29, P2, parallelizable] Author tests/stress/test_tracker_fetch_stress_chaos.py (download-proxy tracker-fetch + cookie-auth path, overlapping the already-fixed SSRF surface) with real §11.4.85 fault injection (process-kill, network-fault, resource-exhaustion — not just concurrency). Verification: each new chaos test is negation-proven (fails without the fault-tolerance code, passes with it), captured evidence per §11.4.5/§11.4.69/§11.4.107. Priority: P2. Composes with GA-20.

BOB-095 — RD2-30: Author tests/ stress/ test_scheduler_hooks_sse_stress_chaos. py for Go-side triangle

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-30, P2, parallelizable] Author tests/stress/test_scheduler_hooks_sse_stress_chaos.py covering the Go-side internal/service/sse_broker.go + internal/api/hooks.go + internal/api/scheduler_api.go triangle. Verification: each new chaos test is negation-proven (fails without the fault-tolerance code, passes with it), captured evidence per §11.4.5/§11.4.69/§11.4.107. Priority: P2. Composes with GA-20.

BOB-096 — RD2-31: Extend qBitTorrent-go jackett_db_test.go with real process-kill/resource-exhaustion fault injection

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-31, P2] Extend qBitTorrent-go/tests/integration/jackett_db_test.go with real process-kill / resource-exhaustion fault injection (currently concurrency-only per GA-20). Priority: P2.

BOB-097 — RD2-32: Author DDoS-class coverage for exposed download-proxy/merge endpoints (canonical impl of RD2-07)

Status: Queued **Type:** Task **Severity:** Low

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-32, P3] Author DDoS-class coverage (RD2-07) for the exposed download-proxy/merge endpoints. Priority: P3.

BOB-098 — RD2-34: Parametrize 20 hardcoded /Volumes/T7 paths in helixqa banks with PROJECT_ROOT (closes GA-23)

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-34, P1 — closes GA-23] submodules/helixqa/banks/.yaml — *parametrize the 20 hardcoded /Volumes/T7/Projects/Boba/... paths with \$PROJECT_ROOT (GA-23) — lands in the submodule per \$11.4.28 decoupling, not boba own tree. GA-23 evidence: grep -rn /Volumes/T7 submodules/helixqa/banks/.yaml returns 20 hits.* Priority: P1.

BOB-099 — RD2-36: Fix guard-forbidden-commands.sh substring-match false-positive class + add const033-poweroff-signal-triage carrier to EXCLUDE_PATHS

Status: Queued **Type:** Bug **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-36, P2] Fix guard-forbidden-commands.sh substring-match false-positive class (RD2-01) — word-boundary/token-aware

matching, not another EXCLUDE_PATHS band-aid; also add the newly-discovered docs/incidents/2026-08-07-const033-poweroff-signal-triage.md carrier to EXCLUDE_PATHS as an immediate stopgap (GA-24 residual). Priority: P2. Composes with RD2-01, GA-24 residual, and RD2-41 (same class in mutation-marker scan).

BOB-100 — RD2-39: Bump submodules/jackett one commit (canonical impl of RD2-09)

Status: Queued **Type:** Task **Severity:** Low

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-39, P3] Bump submodules/jackett one commit (RD2-09). Priority: P3.

BOB-101 — GA-19/RW-09: Is -profile go parity still a release goal? (gates RW-10..13) — OPERATOR-DECISION

Status: Operator-blocked **Type:** Task **Operator-Block-Details:** WHAT: OPERATOR must decide whether Go -profile go parity remains a v1.0.0 release goal; gates downstream RW-10..13. WHY: §11.4.66 forbids autonomous decision on release scope; §11.4.122 forbids silent removal of an existing capability. UNBLOCK: [A] operator states YES — parity remains a release goal (schedule RW-10..13 work) · [B] operator states NO — mark parity Obsolete/superseded per §11.4.90 (reason=feature-removed with operator citation, §11.4.122) WHO: Operator **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md GA-19/RW-09, OPERATOR-DECISION] Confirmed unchanged: zero time.Ticker anywhere in qBitTorrent-go, no enricher package, zero exec.Command fan-out. Is -profile go parity still a release goal? Gates RW-10..13. Surfaced only, not auto-executed, per §11.4.66. Priority: OPERATOR-DECISION.

BOB-102 — RW-05: LAN-exposure threat model — bind tunnel 127.0.0.1 or keep 0.0.0.0? — OPERATOR-DECISION

Status: Operator-blocked **Type:** Task **Operator-Block-Details:** WHAT: OPERATOR must decide LAN-exposure posture: bind tunnel to 127.0.0.1 (loopback-only) OR keep 0.0.0.0 (LAN-reachable, relying on post-RD2-22 complete auth coverage). WHY:

This is a security-posture policy call the agent cannot make autonomously (§11.4.66 operator-decision, §11.4.101 high-blast-radius reversible-only-if-explicit). UNBLOCK: [A] operator states BIND 127.0.0.1 (loopback-only; agent will change compose/service binding + regression-test LAN unreachability) · [B] operator states KEEP 0.0.0.0 (rely on §RD2-22 completed auth coverage; agent will add a permanent §11.4.135 LAN-auth regression guard) WHO: Operator **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RW-05 ungrouped, OPERATOR-DECISION] LAN-exposure threat model — bind the tunnel 127.0.0.1 or keep 0.0.0.0 with the now-complete auth coverage (post Root Cause 3)? Priority: OPERATOR-DECISION.

BOB-104 — §11.4.238 followup: CodeGraph 1.5.0 nested-.gitignore regression challenge

Status: In progress **Type:** Task **Severity:** Medium **Created-By:** Claude

Coverage-escape followup (docs/QA_DISCOVERY_LEDGER.md, BOB-075 agent-code-reading finding, commit e6162f7): CodeGraph 1.5.0 (up from documented 0.9.9) walked into nested-.gitignore-excluded frontend/node_modules and extension/node_modules (32,260 files / 514,456 nodes vs the 2026-06-06 baseline of 509 files / 8,906 nodes) instead of honoring frontend/.gitignore + extension/.gitignore per git check-ignore -v. Author a challenge (challenges/scripts/codegraph_gitignore_honor_challenge.sh or equivalent, e.g. wrapping 'codegraph doctor -sniff-gitignore-honor' if that subcommand exists, else a real re-index + file-count assertion) with §11.4.115 RED_MODE polarity: RED_MODE=1 reproduces the blowup against the live nested-.gitignore tree, RED_MODE=0 asserts the resync stays within the documented baseline order of magnitude. Full evidence: docs/codegraph/Status.md lines 91-118. UPSTREAM FILED 2026-08-18: real upstream repo is github.com/colbymchenry/codegraph (npm package @colbymchenry/codegraph, confirmed via package.json + gh repo view; NOT vasic-digital/codegraph, correcting this ledger's original assumption). Issue: <https://github.com/colbymchenry/codegraph/issues/1567> (full reproduction evidence: git check-ignore -v re-verified first-hand; two bounded synthetic repro attempts up to 63 nested .gitignore files / 960 files against the actual installed v1.5.0 binary did NOT reproduce the blowup in isolation - honest negative result, root cause not pinned to a specific line in either scanning implementation). Draft PR (diagnostic only, not a behavioral fix): <https://github.com/colbymchenry/codegraph/pull/1568> - adds a logDebug() call so a future report can confirm

which of the two independent gitignore-respecting code paths (git-ls-files-based vs filesystem-walk fallback) actually ran. Both PRs/issue filed via SSH (Hard Stop #2) from a fork at github.com/milos85vasic/codegraph. Status: In-progress pending upstream maintainer triage/merge – boba-side closure of this item still needs the RED/GREEN challenge script (§11.4.115) authored per the original scope, independent of upstream’s response.

BOB-105 — §11.4.238 followup: mechanical §11.4.227(B) anchor-block- integrity check

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

Coverage-escape followup (docs/QA_DISCOVERY_LEDGER.md, meta-defect found during this session’s §11.4.140/141 collision resolution, boba commit 136a22c + constitution commits 5ed8c80..e5f2891): §11.4.227(B) anchor-block-integrity (exactly-once heading per anchor per governance file, byte-identical lockstep across mirrors, no anchor-number collision) is fully specified in constitution prose but its own named gate CM-ANCHOR-BLOCK-INTEGRITY is explicitly ‘gate-code = separate work item, NOT claimed shipped’ — the §11.4.140/§11.4.141 double-mandate collision this session’s Phase 0 resolved was verified entirely BY HAND (md5 of moved anchor bodies, manual grep counts pasted into a commit message), not by a runnable script. Propose + author a challenge (in this project’s challenges/scripts/, since this project cannot edit the constitution submodule’s own gate-code from this ticket) that greps every governance file (CLAUDE.md/AGENTS.md/QWEN.md/GEMINI.md/Constitution.md, wherever this project’s constitution submodule checkout exposes them) for ‘### §11.4.NNN’ heading occurrences and FAILs on >1-per-file-per-anchor or on two DIFFERENT anchor bodies sharing one NNN, per §11.4.227(B). Do NOT touch constitution-owned files while implementing this — the check reads them read-only.

BOB-106 — §11.4.238 followup: §11.4.84 quiescence-check helper for the unattributed auto-commit path

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

Coverage-escape followup (docs/QA_DISCOVERY_LEDGER.md RD2-00/BOB-068 entry, docs/GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-00): 20 bare 'Auto-commit' commits exist in this repo's git history (confirmed via git log -oneline -all -grep, e.g. 54e313f/9c8f684/743097a/de9270b/1c36777/41179c2/7c529ca) with no ATM-NNN reference and no TDD trail, landing via ordinary git pull fast-forward from a second session/host with push access to the same remotes (mismatched commit timezone vs the investigating host, per RD2-00 Update). §11.4.84 working-tree-quiescence has no mechanical guard on this path — no gate flags a commit reaching main with a bare/templated message and no ticket citation. Author a §11.4.84 quiescence-check helper (e.g. challenges/scripts/no_unattributed_autocommit_challenge.sh) that scans the commit range since the last known-good release tag and FAILs on any commit message matching a closed bare/templated pattern (e.g. ^Auto-commit\$, ^sync:) with no ATM-NNN or task/PR reference, wired into scripts/pre_build_verification.sh or the §11.4.234 commit-push-all.sh entryptpoint. BOB-068 (RD2-00) remains the tracking item for identifying/stopping the source; this item is specifically the new automated CHECK.

BOB-107 — §11.4.238 followup: pre-dispatch existence check for subagent task-brief source inputs

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

Coverage-escape followup (docs/QA_DISCOVERY_LEDGER.md SCRATCH-LOSS-2026-08-18 entry, evidence: .superpowers/sdd/task-phase1a-report.md line 21): the Phase 1a subagent (a1cc331d) discovered at task start that 5 source files its brief named as required reads (curriculum_amendment_plan_v1.md, ai_curriculum_modules_27_35_extracted.md, and 3 curriculum_analysis_modules_*.md gap analyses) were absent from the session scratchpad — root cause per that subagent's own investigation: the prior producer subagent (ae59171f) hit its session rate limit (a §11.4.147(e) API-quota crash) before writing them. curriculum_amendment_plan_v1.md remains absent from the live scratchpad as of this session's re-check. No mechanical check verifies a task brief's declared input files exist and are non-empty before the downstream consumer subagent is dispatched. Author a pre-dispatch precondition helper (project-side orchestration tooling, e.g. a small script the conductor runs before Task/Agent dispatch when a brief names required input paths) that fails closed with an actionable 'missing input, respawn the

producer' message rather than silently letting a downstream agent proceed on absent evidence and fabricate content unsupported by its named sources.

BOB-109 — BOB-074 followup: scaling-class test coverage absent from mandated test-type matrix

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

docs/testing/test_type_matrix.md's §11.4.27 test-type audit found zero scaling-class coverage anywhere in the tree (no scaling-tagged directory, test file, or HelixQA bank distinguishes growing-dataset/tracker-count/concurrent-user scale-out from stress-under-burst). Scope at least one scaling dimension, e.g. tracker-count scale-out in merge search against challenges/helixqa-banks/boba-services.yaml's tracker set, or the qbittorrent-proxy-go -profile go swap, with a real measured baseline.

BOB-110 — BOB-074 followup: UX-class test coverage (accessibility/usability) absent

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

docs/testing/test_type_matrix.md's §11.4.27 test-type audit found UI functional coverage (Vitest + Playwright) but nothing framed around usability/accessibility/UX outcomes specifically. Scope an axe-core or equivalent accessibility pass over the Angular frontend, covering WCAG checks, keyboard-nav coverage, and screen-reader labeling.

BOB-111 — BOB-074 followup: configure real rate limiting for boba's 3 public HTTP endpoints

Status: Queued **Type:** Task **Severity:** High **Created-By:** Claude

Source inspection across the whole stack (2026-08-18) verified no rate-limit mechanism exists for :7185 (qBittorrent WebUI proxy), :7187 (merge search service), or :7189 (boba-jackett) - no slowapi/limiter/throttle import in download-proxy/src/, no rate-limit middleware in qBitTorrent-go/internal/middleware/ (only

cors.go + logging.go) or internal/jackettapi/ (only auth/cors middleware tests, no rate middleware), and no nginx/reverse-proxy service in docker-compose.yml. Candidate remediations documented in docs/testing/ddos_resilience.md: nginx-in-container reverse proxy with limit_req_zone (most portable, adds a container); a FastAPI slowapi dependency for the merge-search service; a Gin rate-limit middleware for boba-jackett following the existing internal/middleware/ pattern. qBittorrent's own WebUI bandwidth-shaping settings were NOT verified to cover request-rate (only bandwidth) - do not assume they close this gap without checking.

BOB-112 — boba-jackett /healthz amplifies under cold-start concurrent burst via uncached Jackett.GetCatalog() call

Status: Queued **Type:** Bug **Severity:** High **Created-By:** Claude

qBitTorrent-go/internal/jackettapi/health.go:60-63 - HandleHealth makes a synchronous, uncached call to Jackett.GetCatalog() on every single hit to /healthz, with no cache, no distinct timeout, and no circuit breaker. Measured evidence (docs/testing/ddos_resilience.md Findings, 2026-08-18, RED_MODE=0, three independent live runs): up to 98/150 (65%) of health-check requests timed out at 3s under a modest cold-start concurrent burst (10-50 concurrency), recovering to <50ms/request once the burst subsided. This is a genuine self-inflicted DDoS amplification vector: an attacker or a mis-configured monitoring probe hitting /healthz too aggressively can make the Jackett-management API's own health surface appear down without ever touching Jackett itself. Recommended fixes: cache the Jackett liveness signal with a short TTL refreshed by a background ticker; add a tight timeout/circuit-breaker around the GetCatalog call so /healthz itself never blocks past ~250-500ms regardless of Jackett's state. Discovered + scaffolded by BOB-074 (commit ae2b5cb, challenges/scripts/ddos_resilience_challenge.sh); tracked as SDD session task #64 in .superpowers/sdd/progress.md prior to this DB filing - this item is the canonical, tracked workable-items record for that reference (§11.4.93 SSoT, §11.4.214 recurrence-links-not-mints: no prior BOB-NNN existed for this defect, verified by title/description search before minting).

BOB-113 — BOB-074 followup: add wrk to dev tooling for DDoS/load challenges

Status: Queued **Type:** Task **Severity:** Low **Created-By:** Claude

wrk is not installed on the development host; ApacheBench (ab) is, and was used as the primary DDoS-resilience-challenge load tool per the task brief's 'wrk or ab' fallback path, with a curl-parallel-loop as the actual per-status-code assertion mechanism (more precise than ab's aggregate summary) and ab's own output pasted as supplementary real-tool evidence. Install wrk (e.g. via the project's setup.sh or a documented apt/build-from-source recipe) so future load/DDoS challenges have a modern HTTP benchmarking tool with latency-percentile histograms available out of the box, matching the brief's stated primary preference.

BOB-114 — BOB-074 followup: self-validation golden-bad fixture for the rate-limit detector

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

challenges/scripts/ddos_resilience_challenge.sh's -self-validate mode currently only ships a golden-bad fixture for the crash-resistance detector (per §11.4.107(10)/§11.4.201). The rate-limiting detector has no matching golden-bad fixture proving it would actually FAIL a synthetic no-rate-limit-enforced artifact, so an unvalidated rate-limit detector could silently pass a broken/absent rate-limit deployment. Add a synthetic fixture (e.g. a stub server that never returns 429/503 under burst) and assert the detector correctly reports the absence, closing the self-validation gap for this detector class.

BOB-117 — rutracker login diag still uses forbidden §11.4.6 'likely' vocabulary + wrong error_type (unfixed sibling of nnmclub fix)

Status: Queued **Type:** Bug **Severity:** High **Created-By:** AI

search.py:1329 (_search_rutracker password-login path) still classifies a no-session-cookie login result as error_type=auth_failure with message 'rutracker login returned no session cookie — likely CAPTCHA wall or credential failure'. The adjacent code comment (lines 1244-1248, same function)

already states as FACT that ‘rutracker gates login.php behind a CAPTCHA when logins spike... the CAPTCHA is the real blocker’ — yet the error classification still hedges with forbidden §11.4.6 vocabulary (likely) and reports the WRONG error_type (auth_failure instead of upstream_captcha), unlike the exact same class of bug already fixed for nnmclub per docs/qa/nnmclub-login-diagnosis-20260616.md (which replaced ‘likely credential failure’ with a FACT-based upstream_captcha classification + explanatory comment explicitly citing ‘This is NOT likely a credential failure’). The rutracker code path could disambiguate the same way nnmclub’s fix does (or by reading the login response body for a captcha marker, mirroring the existing pattern used ~20 lines later in the SAME function for the search response) but currently discards that opportunity and ships a guess. Found during a §11.4.6 bluff audit (docs/qa/task-bluff-audit/). Fix direction: apply the same upstream_captcha reclassification + FACT-based message used for nnmclub, or add a captcha-marker check on the login response before falling back to a hedged message.

BOB-118 — README.md python-tests badge claims 585 passing; pytest -collect-only measures 5235 (9x stale/wrong)

Status: Queued **Type:** Bug **Severity:** High **Created-By:** AI

README.md line 28 ships a green/success-colored shields.io badge ‘python tests-585 passing’, and line 333 repeats ‘585+ tests’. A real, non-destructive verification this session (timeout 90 nice -n 19 ionice -c 3 .venv/bin/python -m pytest tests/ -collect-only -q) collected 5235 tests, not 585 — roughly 9x higher, so the badge is either wildly stale or was never machine-derived (violates §11.4.259 CM-BADGE-MACHINE-DERIVED-SOURCE + §11.4.6 no-guessing). The README’s own §11.4.259 provenance footnote (lines 40-57) explicitly discloses that the tests/vitest/plugins/merge/scan/license badges were ‘carried forward unverified this session (out of this task’s scope)’ and directs readers to docs/TESTING.md ‘for the current authoritative per-suite counts’ — but docs/TESTING.md contains NO occurrence of ‘585’ or ‘182’ anywhere, so the cited authoritative source cannot actually corroborate the badge. The frontend vitest badge (182 passing) is also suspect: a source-level grep of it()/test() calls in frontend/src/**/*.spec.ts counts ~371, roughly 2x the claimed 182 (lower-confidence proxy than the pytest collect-only count, but consistent with the same staleness pattern). Found during a §11.4.6 bluff audit (docs/qa/task-bluff-audit/). Fix direction: wire a real badge-computer (README already flags this as an owed §11.4.197 gate-

code item) that regenerates the badge from a real collect/run count, and add the actual per-suite counts to docs/TESTING.md so the cited authoritative source is real.

**BOB-119 — docs/
MERGE_SEARCH_DIAGNOSTICS.md
states ENABLE_DEAD_TRACKERS
default=1; actual code + compose
default=0 (contradicts sibling doc)**

Status: Queued **Type:** Bug **Severity:** Medium **Created-By:** AI

docs/MERGE_SEARCH_DIAGNOSTICS.md line ~128 states 'Default: 1 (all trackers exposed; dead ones filtered by DEAD_PUBLIC_TRACKERS)' for ENABLE_DEAD_TRACKERS. This is factually wrong: download-proxy/src/merge_service/search.py:1032 reads os.getenv('ENABLE_DEAD_TRACKERS', '0') (default string '0') and docker-compose.yml:177 sets ENABLE_DEAD_TRACKERS=\0 (also default 0). The sibling doc docs/DEAD_TRACKERS_EXPLAINED.md correctly states 'With ENABLE_DEAD_TRACKERS=0 (default): 24 public trackers active, 14 excluded' — confirmed against the current DEAD_PUBLIC_TRACKERS frozenset (14 entries, names match exactly). MERGE_SEARCH_DIAGNOSTICS.md's stated default is the one document that disagrees with the source of truth, and could mislead an operator into believing dead trackers are shown to end users by default when they are actually filtered out by default. Found during a §11.4.6 bluff audit (docs/qa/task-bluff-audit/). Fix direction: correct 'Default: 1' to 'Default: 0' in MERGE_SEARCH_DIAGNOSTICS.md to match search.py/docker-compose.yml.